

Course Code	Name of the Course			
216U03C501	Digital Communication			
Teaching Scheme	TH	P	TUT	Total
	03	--	--	03
Credits Assigned	03	--	--	03
Evaluation Scheme	Marks			
	LAB/TUT CA	CA (TH)		ESE
		IA	ISE	
	--	20	30	50
100				

Course pre-requisites:

- Communication Systems
- Basics of probability theory

Course Objectives:

The objective of the course is to impart knowledge of digital communication systems with emphasis on digital modulation techniques. It helps in understanding the basic building blocks of a digital communication system and analyse the signal flow in the system. The course analyses error performance of a digital communication system. It introduces the concept and basics of information theory and the basics of source and channel coding /decoding.

Course Outcomes (CO):

At the end of successful completion of the course the student will be able to

- CO1.** Apply information theory in source coding
- CO2.** Analyse distortion less baseband transmission and reception of signal along with waveform coding techniques
- CO3.** Analyse different band pass digital transmission and reception of signal.
- CO4.** Apply concept of sampling for reliable communication.
- CO5.** Apply different methods of error control coding and schemes required for reliable transmission of digital communication.

Detailed Syllabus

Module No.	Unit No.	Contents	Hrs	CO
1	Sampling Process and types of sampling		02	CO1
	1.1	Sampling theorem for low pass & band pass signal, Ideal sampling process, Spectrum of sampled signal, Reconstruction of sampled signal and amplitude response of reconstruction filter.(No derivation)		
	1.2	Signal distortion in sampling, signal-to-distortion Ratio, practical aspect of sampling, flat top sampling, spectrum of flat top sampled signal.		
2	Information Theory , Source coding and Channel capacity		08	CO2
	2.1	Basic Signal Processing Operations in Digital Communications, Uncertainty, Information, and Entropy		
	2.2	Source Coding Theorem, Shannon Fano source coding, Huffman source coding , Lempel –Ziv		
	2.3	Channel capacity, Channel coding theorem, Channel Capacity Theorem		
3	Baseband Modulation and Demodulation Techniques		08	CO3
	3.1	Signal representation: GRAM-SCHMIDT Orthogonalization procedure, Geometric representation of signals		
	3.2	Baseband data formats and their power spectral density: NRZ unipolar, NRZ polar, NRZ bipolar, Manchester, M-ary PAM signals (derivations not expected)		
	3.3	Baseband signal detection and estimation: Maximum likelihood detection and calculation of its probability of error, matched filter receiver , correlation receiver		
	3.4	Nyquist criteria for distortion less baseband transmission: Intersymbol interference(ISI),correlative coding- Duobinary signaling, eye pattern,equalization for efficient baseband data transmission		
		#Self learning topic: non-coherent digital communication systems		
4	Bandpass Modulation and Demodulation		12	CO4
	4.1	Bandpass digital transmitter and receiver model, digital modulation schemes		
	4.2	Generation, detection, spectrum, bandwidth efficiency, signal space diagram and probability of error analysis of Phase Shift Keying: BPSK, QPSK, DPSK, DEPSK Amplitude Shift Keying: BASK Frequency Shift Keying: BFSK, Quadrature Amplitude Modulation (QAM), Minimum Shift Keying (MSK) , Gaussian MSK		

5	Error Control Coding		
	5.1	Linear block codes, Hamming code, generator matrix, systematic linear block codes, parity check matrix, syndrome testing, error correction and decoder implementation	15
	5.2	Cyclic codes: Algebraic structure of cyclic codes, binary cyclic code properties, encoding in systematic form, circuits for dividing polynomials, systematic encoding with shift register and error detection.	
	5.3	Convolutional coding and decoding, State and Trellis Diagram, Viterbi algorithm Overview of Turbo code, Low Density Parity Check codes	
		Total	45

Students should prepare self-learning topics on their own. Self – learning topics will enable students to gain extended knowledge of the topic. Assessment of these topics may be included in IA and or laboratory experiments.

Reference Books*

Sr No	Name/s of Author/s	Title of Book	Publisher	Edition/Year
1	Haykin Simon	<i>Digital Communication Systems</i>	John Wiley and Sons, India	1 st edition 2013
2	Taub Schilling	<i>Principles of Communication systems</i>	Tata McGraw Hill, India	4 th Edition 2013
3	T. L. Singal	<i>Digital Communication</i>	McGraw Hill Education , India	1 st Edition 2015
4	Bernard Sklar and Pabitra Kumar Ray	<i>Digital communications</i>	Pearson Education	2 nd Edition 2014
5	Ranjan Bose	<i>Information theory coding and cryptography</i>	McGraw Hill Education , India	3 rd Edition 2016